

DAF26TX06-NV514 - Space Domain Operational Environment Assessment (Technical Area 2)

USAF / USSF | STTR | Release 6 CSO | Opens Sep 23, 2026 (TPOC direct contact ends) | DSIP Q&A closes Oct 7, 12:00 ET | Proposals due Oct 21, 12:00 ET

Recommendation: PURSUE TA2 only, with an astrodynamics research partner. TA2 is a data-fusion / analytics / decision-support software design problem; TA1 (new sensing phenomenologies) is out of reach. Small award, 10-page limit, ITAR. Confidence MEDIUM.

Phase I award	\$175,000, 6 months
Proposal format	Technical volume 10 pages (DAF STTR CSO index table)
Compliance	ITAR/EAR clause. CMMC Level 2 (Self). STTR: RI >= 30%. 'GFE will not be provided.'
Phase I (TA2)	Design a workable, scalable prototype concept; identify data sources; define metrics and Phase II success criteria
Phase II (TA2)	Prototype software assessing space operations from multi-INT data, shown with modeling, simulation and real-world data

What they want

[TOPIC TEXT] TA2: "Design a concept for a workable and scalable prototype to address the basic capabilities and technical feasibility of the stated objective as well as data and information sources and the timeliness of their solution. Identify non-traditional phenomenology and data sources. Define a set of metrics and success criteria for a Phase 2 prototype."

The problem is operations assessment: compare forecasted outcomes to actual events to judge whether a space task achieved its effect, in a domain where operators cannot directly observe effects or adversary reaction.

Phase II success criteria: a demonstrated software prototype measured against metrics and a CONOPS use case; outputs that integrate into operational workflows with minimal user burden; a Phase III path with customers and funding.

Phase III explicitly includes maturing the software into "a commercial product for commercial SDA" and providing "health and operational status of commercial satellites and their services in relation to adversaries' actions."

How we would build it

- **Data layer:** ingest public catalogs (element sets, conjunction messages), space-weather feeds, launch notices, and open-source reporting as a proxy for multi-INT; normalize to an event/object graph with provenance.
- **Pattern-of-life models:** per-object maneuver and RF/optical behavior baselines; anomaly detection for unexpected maneuvers, station-keeping breaks, proximity operations.
- **Assessment engine:** represent a planned task with measures of performance/effectiveness, forecast expected observables, then score observed vs. forecast - the 'did it work?' logic the topic is really asking for.
- **Timeliness metrics:** detection lead time, time-to-assessment, false-alarm rate - defined in Phase I, measured in Phase II.
- **Operator view:** a lightweight assessment dashboard that fits into existing workflows (the same pattern as the Release 5 NAVWAR APNT topic).

Resources we would use

- Public catalog data (space-track.org, Celestrak), conjunction data messages, space-weather services.
- Open astrodynamics libraries for propagation and maneuver detection; the AMOS conference papers cited in the topic (multi-phenomenology characterization with RL; passive-RF cislunar tracking) for the state of the art.
- Joint Publications 3-14, 5-0, 3-60 and the Space Capstone Publication (cited) for assessment doctrine and vocabulary.
- RI partner: a university astrodynamics / space-situational-awareness group (covers orbital mechanics credibility).
- No hardware; no GFE will be provided, so the data plan must be entirely public/commercial.

Bottlenecks and risks

- **Data realism:** real multi-INT is classified; the proposal must argue convincingly that public/commercial data suffices to prove the assessment logic, and that classified feeds slot in later.
- **Commercial incumbents:** commercial SDA analytics firms already sell catalog/conjunction products; our lane must be assessment and adversary-intent reasoning, not another catalog.
- **10-page limit and STTR partner:** both constrain how much can be shown; a strong figure of the assessment loop is essential.
- **ITAR staffing.**
- **Scope creep into TA1:** stay out of new-sensor territory; reviewers will score TA2 on the analytics design.

Decision factors

- Fit to team: data integration, anomaly detection, dashboards - strong; astrodynamics via the RI.
- Dual use: explicitly stated by the sponsor (commercial SDA product, commercial satellite health vs. adversary action).
- Decisive pre-proposal experiment: from public element-set histories, detect a set of known, publicly reported maneuvers and report lead time and false alarms - a concrete TA2 metric to put in the proposal.

Before Oct 7 (Q&A): ask whether TA2-only proposals are welcome, what 'non-traditional data sources' the sponsor has in mind for TA2, and whether any unclassified operational assessment examples can be shared.

Tags: [TOPIC TEXT] = quoted from the solicitation; [Q&A] = government answer on DSIP; [INFERRED] = our judgment, not verified by search. Generated 2026-09-20 from release6_triage.md / triage_results.md / sitis_qa.md.